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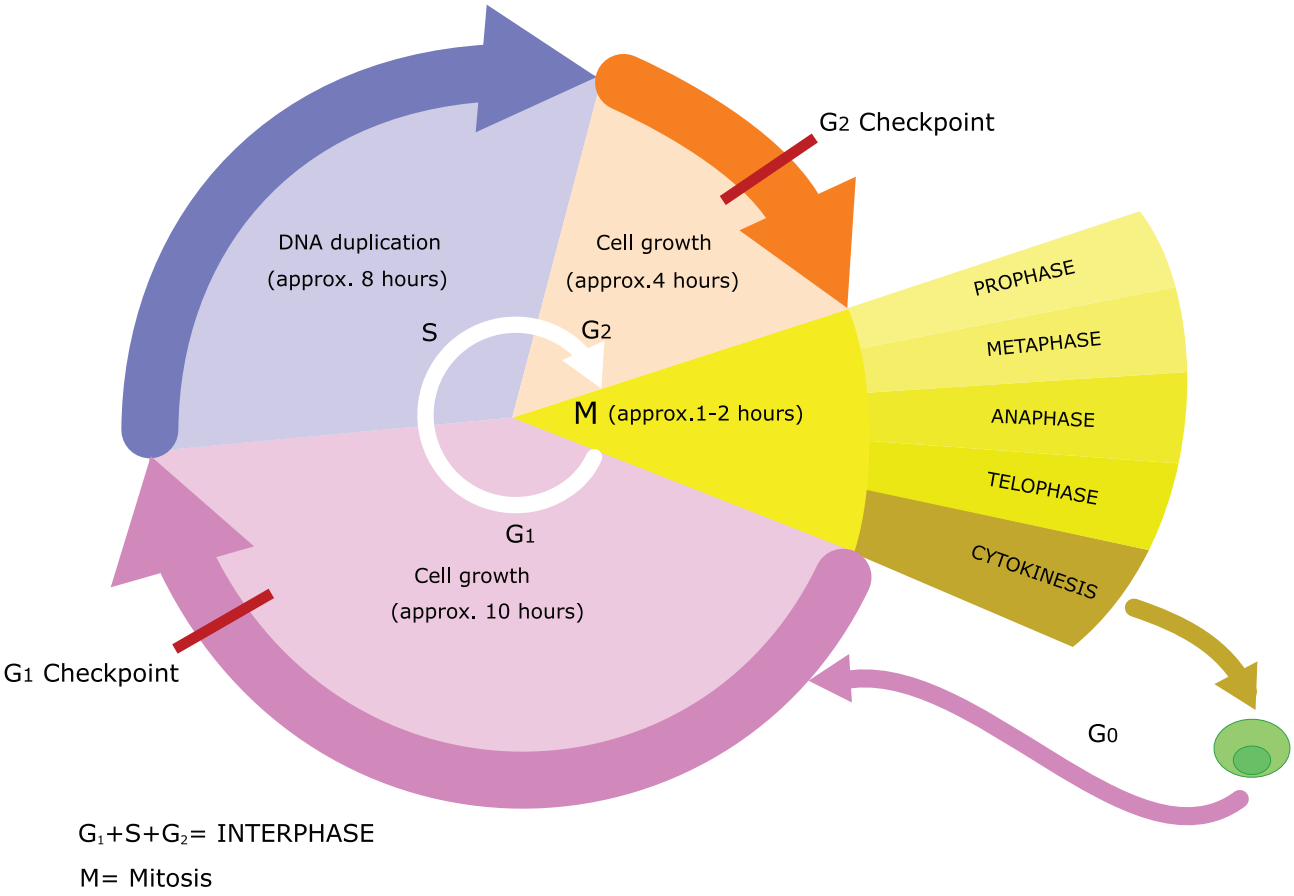


Figure 7.1: Cell cycle

the cell cycle. The checkpoint called the **restriction point** at the end of G_1, determines a cells fate whether it will continue in the cell cycle and divide or enter a stage called **G_0** as a quiescent stage and probably as specified cell or die. Cells are arrested in G_1 due to:

- Nutrient deprivation
- Lack of growth factors or density dependant inhibition
- Undergo metabolic changes and enter into G_0 state.

Biochemicals inside cells activates the cell division. The proteins called **kinases** and cyclins activate genes and their proteins to perform cell division. Cyclins act as major checkpoint which operates in G_1 to determine whether or not a cell divides.



Dolly

Since the DNA of cells in G_0, do not replicate. The researcher are able to fuse the donor cells from a sheep's mammary glands into G_0 state by culturing in the nutrient free state. The G_0 donor nucleus synchronised with cytoplasm of the recipient egg, which developed into the clone Dolly.

7.2.4 G_0 Phase

Some cells exit G_1 and enters a quiescent stage called **G_0**, where the cells remain metabolically active without proliferation. Cells can exist for long periods in G_0 phase. In G_0 cells cease growth with reduced rate of RNA and protein synthesis. The G_0 phase is